



Sec: 5C-60

D-Block elements

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SINGLE ANSWER TYPE:

- Arrange the following ions in their magnetic moment :
(i) V^{4+} (ii) Mn^{4+} (iii) Fe^{3+} (iv) Ni^{2+}
1) $ii > iii > i > iv$ 2) $iii > iv > ii > i$ 3) $iii > ii > iv > i$ 4) $i < iv < iii < ii$
- Which of the following is not a consequence of Lanthanoid contraction ?
1) 5d series elements have a higher IE_1 than 3d or 4d series
2) Zr and Hf have a comparable size
3) Zr and Hf occurs together in the earth crust in their minerals
4) High density of the period 6 elements
- Among V, Mn, Cr and Fe, which one is expected to have the highest third ionization enthalpy ?
1) V 2) Mn 3) Cr 4) Fe
- Of the following outer electronic configurations of atoms, the highest oxidation state is achieved by which one of them ?
1) $(n-1)d^8ns^2$ 2) $(n-1)d^5ns^1$ 3) $(n-1)d^3ns^2$ 4) $(n-1)d^5ns^2$
- The spin only magnetic moment of Ni^{2+} in aqueous solution would be [in units of Bohr magneton].
1) 2.84 2) 4.90 3) 0 4) 1.73
- Gun metal is an alloy of
1) Cu and Al 2) Cu, Zn and Ni 3) Cu, Sn and Zn 4) Cu, Zn and Al
- Which one of the following is the correct configuration of Fe^{3+}
1) $[Ar]4s^2, 3d^6$ 2) $[Ar]4s^2, 3d^3$ 3) $[Ar]3d^5$ 4) $[Ar]4s^2 3d^4$
- Which of the following pairs of ions are colourless ?
1) Ti^{3+}, Cu^{2+} 2) Sc^{3+}, Zn^{2+} 3) Co^{2+}, Fe^{3+} 4) Ni^{2+}, V^{3+}
- The basic character of the transition metal monoxides follow the order :
1) $CrO > VO > FeO > TiO$ 2) $TiO > FeO > VO > CrO$
3) $TiO > VO > CrO > FeO$ 4) $VO > CrO > TiO > FeO$
- Among the following series of transition ions, the one where all metal ions have $3d^2$ electronic configuration is :
1) $Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+}$ 2) $Ti^+, V^{4+}, Cr^{6+}, Mn^{7+}$
3) $Ti^{4+}, V^{3+}, Cr^{2+}, Mn^{3+}$ 4) $Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}$
- Which one of the following pairs of ions have same paramagnetic moment ?
1) Mn^{2+}, Cu^{2+} 2) Cu^{2+}, Ti^{3+} 3) Ti^{4+}, Cu^{2+} 4) Ti^{3+}, Ni^{2+}
- Which of the following compounds are coloured due to charge transfer spectrum ?
i) $K_2Cr_2O_7$ ii) $KMnO_4$
iii) MnO_2 iv) $[Fe(H_2O)_5 NO]SO_4$
1) all 2) only i and ii 3) i, ii, iii only 4) iii & iv only
- Transition metal ions form coloured compounds due to :
i) d-d transition ii) charge transfer spectrum
iii) their high oxidation states iv) vacant d-orbitals
1) all 2) i, ii, iii 3) i, ii, iv 4) i & ii
- The catalytic activity of transition elements and their compounds is ascribed to :
1) Their chemical reactivity
2) Their magnetic behavior
3) Formation of coloured ions
4) Their ability to adopt multiple oxidation states and their complexing ability

15. Which group contains coloured ions out of :
 A) Cu^+ B) Ti^{4+} C) Co^{2+} D) Fe^{2+}
 1) A, B, C, D 2) C, D 3) B, C 4) A, B
17. The compound of Vanadium has magnetic moment of 1.73 BM. The Vanadium chloride has the formula ?
 1) VCl_2 2) VCl_3 3) VCl_4 4) VCl_5
18. Least paramagnetic property is shown by :
 1) Fe 2) Mn 3) Ni 4) Cu
19. The effect of Lanthanoid contraction is observed in :
 1) Hf 2) At 3) Xe 4) Ac
20. One of the constituents of German silver is
 1) Ag 2) Cu 3) Mg 4) Al
21. The magnetic moment of Gd^{3+} is [in Bohr magneton]
 1) 0 2) 2.84 3) 7.9 4) 6.9
22. Complex forming tendency of transition elements is high, because ;
 1) they have vacant d-orbitals 2) their large size
 3) they form coloured ions 4) they possess high densities
23. The magnetic moment of a transition metal of 3d-series is $6.92\mu_B$. Its electronic configuration will be :
 1) $3d^4, 4s^2$ 2) $3d^5, 4s^1$ 3) $3d^{10}$ 4) $3d^5, 4s^0$
24. The tendency of the transition elements to form coloured compound is attributed to :
 (A) transition of electrons from one atoms to the other.
 (B) transition of electrons from s-orbitals of the outer shells to p-orbitals.
 (C) d-d-transition in last but one shell. (D) none of the reason is correct.
25. Of the ions Zn^{2+} , Ni^{2+} and Cr^{3+} (At. Nos. Zn = 30, Ni = 28, Cr = 24) :
 (A) only Zn^{2+} is colourless and Ni^{2+} and Cr^{3+} are coloured.
 (B) all three are colourless. (C) all three are coloured.
 (D) only Ni^{2+} is coloured and Zn^{2+} and Cr^{3+} are colourless.
26. Which oxide of manganese is most acidic in nature ?
 (A) MnO (B) Mn_2O_7 (C) Mn_2O_3 (D) MnO_2 .
27. In general, the transition elements exhibit their highest oxidation states in their compounds with elements like :
 (A) C (B) S (C) S and P (D) F and O.
28. The maximum oxidation state shown by V(Z = 23), Cr(Z = 24), Co(Z = 27), Sc(Z = 21) are respectively :
 (A) + 5, + 6, + 3, + 3 (B) + 3, + 4, + 5, + 2
 (C) + 5, + 3, + 2, + 1 (D) + 4 in each case.
29. If a non metal is added to the interstitial sites of a metal, then the metal becomes :
 (A) softer (B) less tensile (C) less malleable (D) more ductile.
30. Among the following series of transition metal ions, the one where all metal ions have $3d^2$ electronic configuration is ;
 (A) $\text{Ti}^{3+}, \text{V}^{2+}, \text{Cr}^{3+}, \text{Mn}^{4+}$ (B) $\text{Ti}^+, \text{V}^{4+}, \text{Cr}^{6+}, \text{Mn}^{7+}$
 (C) $\text{Ti}^{2+}, \text{V}^{3+}, \text{Cr}^{2+}, \text{Mn}^{3+}$ (D) $\text{Ti}^{2+}, \text{V}^{3+}, \text{Cr}^{4+}, \text{Mn}^{5+}$.
31. Transition metal with low oxidation number will act as :
 (A) a base (B) an acid (C) both (A) and (B) (D) none of these.
32. S_1 : Mn^{2+} compounds more stable than Fe^{2+} towards oxidation to their +3 state.
 S_2 : Titanium and copper both in the first series of transition metals exhibits +1 oxidation state most frequently.
 S_3 : Cu^+ ion is stable in aqueous solutions.

S₄ : The E^θ value for the $\text{Mn}^{3+} / \text{Mn}^{2+}$ couple is much more positive than that for $\text{Cr}^{3+} / \text{Cr}^{2+}$ or $\text{Fe}^{3+} / \text{Fe}^{2+}$.

(A) T T F T (B) T F F T (C) T F T T (D) F F T F

33. Electronic configuration of a transition element X in +3 oxidation state is $[\text{Ar}]3d^5$. What is its atomic number?

(A) 25 (B) 26 (C) 27 (D) 24

34. Metallic radii of some transition elements are given below. Which of these elements will have highest density?

Element	Fe	Co	Ni	Cu
Metallic radii/pm	126	125	125	128

(A) Fe (B) Ni (C) Co (D) Cu

35. Which of the following is amphoteric oxide?

$\text{Mn}_2\text{O}_7, \text{CrO}_3, \text{Cr}_2\text{O}_3, \text{CrO}, \text{V}_2\text{O}_5, \text{V}_2\text{O}_4$

(A) $\text{V}_2\text{O}_5, \text{Cr}_2\text{O}_3$ (B) $\text{Mn}_2\text{O}_7, \text{CrO}_3$ (C) $\text{CrO}, \text{V}_2\text{O}_5$ (D) $\text{V}_2\text{O}_5, \text{V}_2\text{O}_4$

36. Interstitial compounds are formed when small atoms are trapped inside the crystal lattice of metals. Which of the following is not the characteristic property of interstitial compounds?

(A) They have high melting points in comparison to pure metals.

(B) They are very hard. (C) They retain metallic conductivity (D) They are chemically very reactive.

37. The magnetic moment is associated with its spin angular momentum and orbital angular momentum. Spin only magnetic moment value of Cr^{3+} ion is _____.

(A) 2.87 B.M. (B) 3.87 B.M. (C) 3.47 B.M. (D) 3.57 B.M.

38. Although Zirconium belongs to 4d transition series and Hafnium to 5d transition series even then they show similar physical and chemical properties because _____.

(A) both belong to d-block

(B) both have same number of electrons.

(C) both have similar atomic radius

(D) both belong to the same group of the periodic table.

KEY:

1.	3	2.	4	3.	2	4.	4	5.	1	6.	3	7.	3
8.	2	9.	3	10.	4	11.	4	12.	2	13.	2	14.	3
15.	4	16.	4	17.	2	18.	3	19.	2	20.	4	21.	1
22.	1	23.	2	24.	C	25.	A	26.	B	27.	D	28.	A
29.	C	30.	D	31.	B	32.	B	33.	B	34.	D	35.	A
36.	D	37.	B	38.	C								